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HOLLOMAN AIR FORCE BASE

HOLLOMAN AIR FORCE BASE
INSTRUCTION 15-101

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Weather

WEATHER SUPPORT PROCEDURES

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This instruction implements Air Force Policy Directive (AFPD) 15-1, *Weather Operations*. It establishes the responsibilities and procedures for providing and using weather services at Holloman AFB (HAFB). It provides general information on weather services including weather observations and forecasts, weather warnings and advisories, dissemination of weather information, and reciprocal support. It applies to all 49th Wing (WG) agencies and tenant units described herein. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual 33-363, *Management of Records*, and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS) located at **https://www.my.af.mil/afrims/afrims/afrims/rims.cfm**. Contact supporting records managers as required. Refer recommended changes and questions to the Office of Primary Responsibility (OPR) using Air Force (AF) Form 847, *Recommendation for Change of Publication*, to 54th Operations Support Squadron Weather Flight (54 OSS/OSW), 1148 Rescue Road, Holloman AFB, New Mexico 88330-8029.

SUMMARY OF CHANGES

This document has been completely revised. The chapters contain information on overarching weather principles and weather characterization for Holloman AFB. Guidance specific to the unique operations of weather support will be maintained in the 15-series publications.

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Chapter 1

GENERAL INFORMATION

1.1. Overview.

1.1.1. The 25th Operational Weather Squadron (25 OWS) and 54th Operations Support Squadron Exploitation Unit (54 OSS/OSW) or Weather Flight (WF), herein referred to as the WF, are the official weather information agencies for, and provide weather services to, 49 WG and tenant units assigned to Holloman AFB, New Mexico. Basic concepts and procedures are outlined in the 15-series Air Force and Air Education and Training Command directives and further specified on the Holloman AFB Installation Data Page (IDP).

1.1.2. This instruction establishes requirements and procedures for weather support, which must be coordinated at the local level to meet mission needs. It consolidates weather support requirements and procedures for peacetime operations and eliminates the need for written agreements between the WF and supported units/operations. It does not cover weather support procedures for emergency war operations or certain other special operations and procedures. These are covered in applicable plans and regulations.

1.2. Weather Operations.

1.2.1. The 25 OWS at Davis-Monthan AFB, Arizona provides regional and operational-level weather products and information to Air Force and Army units operating in the western CONUS. The 25 OWS provides regional and operational-level weather products and information, Meteorological Watch (METWATCH), limited flight weather (DD Form 175-1, Flight Weather Briefing) support, and verbal weather briefings.

1.2.2. The WF will focus on providing tactical-level weather products and information needed for routine mission execution and support to the commander during training, wartime, contingency, and exercise operations. The WF provides the Mission Execution Forecast (MEF), base resource protection, and deployment weather information to the 49 WG and tenant units. Primary weather support for transient aircrews will be provided by 25 OWS. The WF may provide transient aircrew support during duty-hours as time permits.

1.3. Assumptions. Adequate resources and communications will be available to execute this instruction and sufficient weather data will be available from various sources on which to base weather operations and production.

1.4. Identifiers, Station Information, and Designators. To maintain current and valid information, all Holloman AFB Identifiers and Designators will be referenced in Department of Defense (DoD) Flight Information Publications/Handbooks and World Meteorological Organization (WMO) designations.

1.5. Duty Priorities. Not all WF tasks can be accomplished simultaneously. Therefore, duty priorities are established to ensure tasks are accomplished in order of importance and publicized to avoid misunderstanding among supported agencies. Duty priorities will ensure timely response to situations under normal conditions. However, the list will not replace situational awareness, mission necessity, and sound judgment. The weather technician may deviate in the best interests of flight safety and/or protection of personnel or property. The weather technician will use **Table 1.1** as a guide for accomplishing duties.

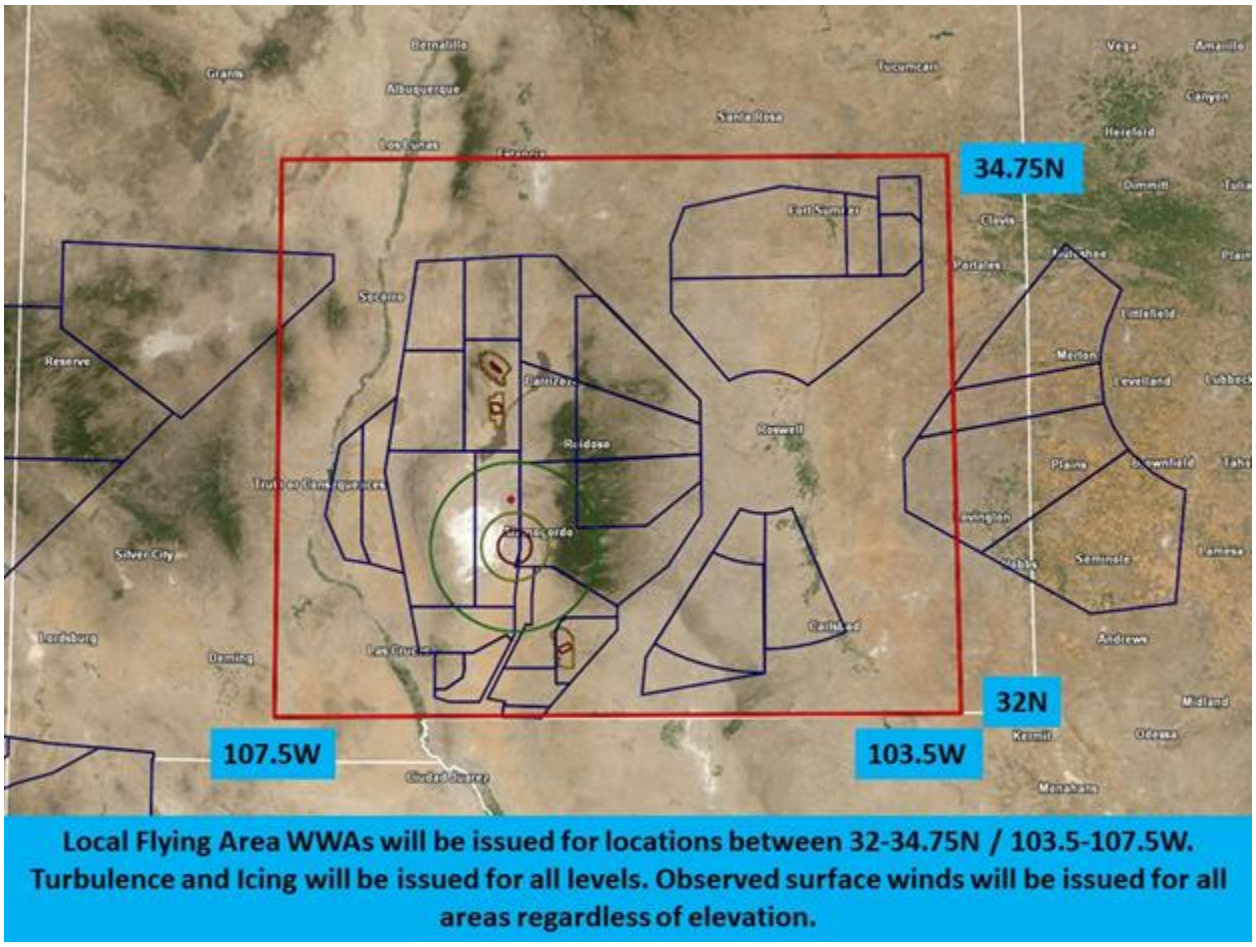
Table 1.1. Holloman AFB Weather Flight Airfield Services Element Duty Priorities

1	Perform Emergency War Order Taskings.
2	Perform Facility Evacuation Procedures.
3	Respond to Aircraft/Ground Emergencies.
4	Respond to and Provide Airborne Aircrew Support through Pilot-to-Metro Services (PMSV).
5	Provide Weather Information for Supervisor of Flying (SOF).
6	Conduct Severe Weather Action Plan (SWAP) operations.
7	Disseminate mission critical Weather Watches, Warnings, and Advisories (WWA). Record and Disseminate Augmented Airfield Weather Observations to include additional and mandatory supplementation criteria for the fixed airfield sensors (FMQ-19). The forecaster will determine which task is to be accomplished first based on the mission and weather phenomena occurring.
8	Disseminate Urgent/Routine Pilot Reports (PIREPs) and Special Air Reports (AIREPs).
9	Produce and disseminate the local Terminal Aerodrome Forecast (TAF).
10	Produce and Disseminate MEF
11	Perform METWATCH and MISSIONWATCH Activities.
12	Provide 'Eyes Forward' and Collaborate with 25 OWS.
13	Provide Transient Aircrew (175-1) and Other Weather Support Briefings.
14	Accomplish Weather Functional Training.
15	Accomplish Administrative and Additional Duty Tasks.

1.6. Geographic Area of Responsibility. The area of responsibility for products and services provided by the WF is the area located within a 5 Nautical Mile (NM) radius of Holloman AFB. The WF will also produce a MEF and MISSIONWATCH for all areas and routes in which 49 WG and tenant flying units are conducting operations.

1.7. Local Flying Area. Figure 1.1 designates the region that local flying area products and services pertain to.

Figure 1.1. Local Flying Area.



1.8. Operating Hours and Contact Information.

1.8.1. Staff services are available from 0730 local time to 1630, Monday through Friday. Exceptions are federal holidays, Air Education Training Command (AETC) family days, and 49th WG and 54 Fighter Group (FG) training days.

1.8.2. Weather technicians are available to provide services at Holloman AFB during airfield hours. A technician will be on duty, as required, prior to scheduled missions in order to prepare and disseminate MEFs.

1.8.3. During non-duty hours, there is one WF member on standby to coordinate communication with the Command Post to determine if the Severe Weather Management Team (SWMT) needs to be activated, to respond to urgent requests, or to troubleshoot meteorological equipment.

1.8.4. **Table 1.2.** provides the contact information for the WF.

Table 1.2. Weather Flight Contact Information.

SECTION	COMMERCIAL NUMBER (**FOR DSN, OMIT AREA CODE)
Airfield Services Element (ASE)	**575-572-3924
Mission Weather Element (MWE) & MQ-9 Weather Operations Center	**575-572-2794
F-16 Weather Operations (314 FS)	**575-572-7399
F-16 Weather Operations (8 FS)	**575-572-8880
F-16 Weather Operations (311 FS)	**575-572-1753
Flight Chief	**575-572-7386
Flight Commander	**575-572-7379
25TH Operational Weather Squadron	520-228-7653
Standby Forecaster via 49 WG Command Post	**575-572-4900
Severe Weather Action Plan Phone	575-430-2024

1.9. Joint Environmental Toolkit (JET). JET is the primary weather data dissemination system on Holloman AFB. All required agencies will have access to the JET portal. It is recommended to have it running at all times in order to automatically receive weather updates as they occur.

1.9.1. Air Traffic Control (ATC) Tower and Radar Approach Control (RAPCON) use the Airfield Advisory System as the primary weather data dissemination system.

1.10. Base Weather Station Evacuation. If the WF is required to evacuate the base weather station (Building 577 at 1148 Rescue Road), the team will relocate to the Alternate Operating Location (AOL), in building 892N. The WF can be contacted at DSN 572-7399 or Commercial (575) 572-7399 at the AOL.

1.10.1. The 25 OWS will assume the following duties until the WF resumes operations:

1.10.1.1. Perform METWATCH and MISSIONWATCH for 49 WG and tenant unit operations and issue all terminal and flying area weather warnings and advisories to the best of their ability. Since the 25 OWS is geographically separated and is limited to cloud-to-ground lightning strike data, it does not have the capability to detect or observe other lightning strike occurrences (e.g., cloud-to-cloud lightning).

1.10.1.2. Produce and update the flight weather support for the 49 WG and tenant units in accordance with AFI 15-128.

1.10.1.3. Assume responsibility for all other flight weather briefings for missions departing Holloman AFB and conduct limited MISSIONWATCH for these flights.

1.10.2. Upon arrival at the AOL, the WF will stand up operations and notify the 25 OWS, ATC, RAPCON, Command Post, and all Holloman AFB flying units that the WF has resumed operations at the AOL.

1.10.3. The weather technician will provide the following services from the AOL:

1.10.3.1. If the fixed airfield weather sensors (FMQ-19) are operative and there is no break in the automated observation report, then no augmented observation will be taken.

1.10.3.2. If the FMQ-19 is inoperative, an augmented observation will be taken within 15 minutes of arriving at the AOL. Surface weather observations will be disseminated using JET as the primary dissemination method. If primary dissemination methods are not available, observations will be disseminated using back-up procedures on the Air Force Weather Web Services (AFW-WEBS) or another Air Force Weather agency.

1.10.3.3. Continue to issue and update MEFs if the capability exists. The MEF will continue to be updated on the WF webpage if possible (SharePoint: **https://usaf.dps.mil/sites/aetc-hmn/Weather/SitePages/Home.aspx**). If the webpage is inaccessible, the MEF will be emailed to the flying units.

1.10.3.4. Provide METWATCH for Holloman AFB from the AOL. Forecast weather warnings, watches, and advisories will be disseminated by the 25 OWS or by phone if JET is non-operational.

1.10.3.5. Brief aircrews and MISSIONWATCH.

1.11. Backup Weather Support. If weather operations at the 25 OWS are interrupted (e.g., power outage, natural disaster), the 25 OWS will inform the WF. Responsibility for other 25 OWS products (e.g., graphical products, JET, and OWS webpage) will be transferred to other agencies as necessary in order to continue weather information flow to Holloman AFB customers.

1.12. Release of Weather Information. Support to non-DoD agencies and the general public will not be provided without the approval of the Wing Public Affairs Office.

Chapter 2

AIRFIELD SERVICES ELEMENT

2.1. Basic Weather Watch (BWW). A BWW will be conducted during periods when the airfield is controlled and during periods when any of the mandatory Automated Meteorological Observation System (AMOS) augmentation criteria is occurring. Weather personnel will check weather conditions at intervals not to exceed **20 minutes** to determine the need for a special observation. Reference **Table 2.1** for augmentation criteria.

2.1.1. The FMQ-19 may be augmented when the airfield is controlled and the weather technician feels augmentation is necessary to accurately support safety of flight.

2.1.2. When a reliable source (ATC personnel, pilots) reports weather conditions different from the last report, weather personnel will recheck the weather and, if required, disseminate a new observation and log out the FMQ-19 if necessary.

Table 2.1. Summary of Mandatory Supplementary Weather Conditions.

Items from AFMAN 15-111
Tornado (+FC) (Notes 1 & 2)
Waterspout (+FC) (Notes 1 & 2)
Funnel Cloud (FC) (Notes 1 & 2)
Freezing Precipitation (FZDZ/FZRA)
Ice Pellets (PL)
Hail (GR)
Sandstorm (SS)/Dust Storm (DS) (Note 3)
Volcanic Ash (VA)
Tower Visibility remark (Note 4)
Items local to Holloman AFB
Blowing dust that causes erroneous ceilings below 3000' and visibility less than 3 miles (Note 5)
Erroneous precipitation reading measurements during prolonged wind events
Thunderstorms are observed on station or w/in 10 NM but not in automated observation. (Note 6)
Notes: 1. The immediate reporting of tornadic activity takes precedence over all other phenomena. 2. Be prepared to supplement whenever a tornado watch is valid or warning has been issued; regardless of airfield closure status. 3. Based on local weather warning criteria; if no warning criteria exists, this is not required. 4. Only required during controlled airfield hours. 5. BLDU - Visibility ≥5/8 SM; DS - Visibilty <5/8 SM but ≥5/16 SM; +DS - Visibilty <5/16 SM 6. TS or VCTS

2.2. Continuous Weather Watch (CWW). The FMQ-19 provides an automatic continuous watch. When augmentation is required and deemed appropriate, weather technicians will monitor weather conditions continuously and perform **no other significant duties**, to determine the need for a SPECI observation and to ensure that the sensor does not need to be augmented.

2.3. Cooperative Weather Watch. The WF will establish a cooperative weather watch with ATC and other appropriate base agencies, as required. Of primary concern is the report of tower visibility different from the prevailing surface visibility, local PIREPs, and any occurrence of previously unreported weather conditions that could affect flight safety or be critical to the safety or efficiency of other local operations and resources. Based on reevaluation of the different weather conditions reported and local policy, the technician will follow the guidance outlined in AFMAN 15-111. See **paragraph 8.3.1.1** of this publication for more specific requirements.

2.4. 'Eyes Forward'. The WF integrates weather radar data, meteorological satellite imagery, lightning detection display, and non-standard weather data systems to create an integrated weather picture for the 25 OWS. 'Eyes forward' yields meaningful meteorological information not contained in coded observations for the 25 OWS and is an integral part of the METWATCH for Holloman AFB. The WF will provide 'eyes forward' to the 25 OWS during WF duty hours by providing significant time-sensitive information to the 25 OWS concerning local area weather patterns and un-forecast changes.

2.5. Terminal Aerodrome Forecast (TAF).

2.5.1. Weather technicians at the Holloman WF will prepare and disseminate a TAF via JET at 0100, 0900, and 1700 Zulu time and cover a 30-hour period. All TAF nomenclature can be found in AFMAN 15-124.

2.5.2. Specification and Amendment Criteria. The TAF Specification and Amendment criteria are outlined in AFMAN 15-129 and the Holloman AFB IDP.

2.5.3. Dissemination. The WF will disseminate the TAF via JET. If JET is inoperative, the TAF will be issued through AFW-WEBS. If AFW-WEBS is inoperative, TAFs will be disseminated verbally to the ATC, SOF, and Command Post when the airfield is open.

2.6. Surface Observations. Note that an "M" indicates mandatory supplemental criteria.

2.6.1. **Observation Criteria** . All observation nomenclature can be found in AFMAN 15-111.

2.6.1.1. **Special Observation Criteria** . Reference **Table 2.2** below for ceiling, visibility, and RVR-specific criteria. Information taken from the DOD Flight Information Publications (FLIPS) and AFMAN 15-111.

2.6.1.1.1. Sky Condition: A layer of clouds or obscuring phenomenon aloft is observed below 700 feet and no layer aloft was reported below this height in the previous Meteorological Aerodrome Report (METAR) or Special (SPECI) observation.

2.6.1.1.2. Wind Shift: The wind direction changes by 45 degrees or more in less than 15 minutes and the wind speed is 10 knots or more throughout the wind shift.

2.6.1.1.3. Squall: A strong wind characterized by a sudden onset in which the wind speed increases by at least 16 knots and is sustained at 22 knots or more for at least 1 minute.

2.6.1.1.4. **(M)** Volcanic Eruption: When eruption is first noted.

2.6.1.1.5. Thunderstorm or Vicinity Thunderstorm: Begins or ends. A SPECI is not required to report the beginning of a new thunderstorm if one is currently being reported.

2.6.1.1.5.1. For reporting purposes, a thunderstorm is considered to have begun and to be occurring "at the station" when (1) thunder is first heard when location is unknown, (2) when hail is falling or lightning is observed at or near the airfield and the local noise level is such that resulting thunder cannot be heard or (3) lightning detection equipment indicates lightning strikes within 5 nautical miles of the airfield.

2.6.1.1.5.2. A thunderstorm is considered to have ended 15 minutes after the last occurrence of any of the above criteria.

2.6.1.1.6. Precipitation:

2.6.1.1.6.1. **(M)** Hail begins or ends, and include the size of the hail in the plain language remarks.

2.6.1.1.6.2. Freezing precipitation begins, ends, or changes in intensity.

2.6.1.1.6.3. Ice pellets begin, end, or change in intensity.

2.6.1.1.6.4. Any other type of precipitation begins or ends. *Note: Except for freezing rain, freezing drizzle, hail, and ice pellets, a SPECI is not required for changes in type or the beginning or ending of one type while another is in progress.

2.6.1.1.7. **(M)** Tornado, Funnel Cloud, or Waterspout:

2.6.1.1.7.1. Is observed, or changes from one to the other.

2.6.1.1.7.2. Disappears from sight or ends.

2.6.1.1.8. **Runway Visual Range (RVR):** Is an instrumentally derived value that represents the horizontal distance that a pilot can see down the runway. The FMQ-19 provides the only RVR measurement for HAFB. If the FMQ-19 no longer provides a measurement, do not report RVR, but include the coding “RVRNO” during back-up procedures. In automated mode, the FMQ-19 automatically records and transmits RVR.

2.6.1.1.8.1. RVR will be determined when prevailing visibility is 1 statute mile (1600 meters) or less and/or RVR is 6000 feet (1500 meters) or less.

2.6.1.1.9. **(M) Tower Visibility:**

2.6.1.1.9.1. When notified by the control tower that tower visibility has decreased to less than or, if below, increased to equal or exceed 1, 2, or 3 statute miles, and the control tower visibility differs from the prevailing visibility.

2.6.1.1.9.2. When notified by the control tower that tower visibility has decreased to less than or, if below, increased to equal or exceed locally developed tower special criteria (if applicable) and the control tower visibility differs from the prevailing visibility.

2.6.1.1.10. **Upon Resumption of Observing Function:** A SPECI will be taken within 15 minutes after the weather technician returns to duty following a break in observing coverage or augmentation at the unit unless a record observation is filed during that 15 minute period.

2.6.1.1.11. **Criteria established locally.**

2.6.1.1.12. **Aircraft Mishap:** (When an AMOS is in supplement or back-up mode). Take an aircraft mishap SPECI immediately following notification or sighting of an aircraft mishap at or near the observing location unless there has been an intervening observation.

2.6.1.1.13. **Miscellaneous:** Any other meteorological situation that, in the observer's opinion, is critical.

Table 2.2. Ceiling, Visibility, and Runway Visual Range Criteria.

Ceiling (CIG) rounded off to reportable values. Reported when it forms or dissipates below, decreases to less than, or, if below, increases to equal or exceed the values listed below.	**Surface visibility (VIS)** as reported in the body of the report decreases to less than, or, if below, increases to equal or exceed the values listed below.	**RVR:** runway visual range for active runway decreases to less than or, if below, increase to equal or exceed.
3,000 (AFMAN 15-111)	**3** (AFMAN 15-111) (Field IFR)	**6000** (AFMAN 15-111)
2,000 (AFMAN 15-111)	**2 ¼** (DoD FLIPS)	**5000** (AFMAN 15-111)
1,500 (AFMAN 15-111) (Field IFR)	**2** (AFMAN 15-111 & DoD Flips)	**4000** (AFMAN 15-111)
1,000 (AFMAN 15-111)	**1 7/8** (DoD FLIPS)	**2400** (AFMAN 15-111)
800 (AFMAN 15-111)	**1 ¾** (DoD FLIPS)	**2000** (AFMAN 15-111)
700 (AFMAN 15-111 & DoD Flips)	**1 ½** (AFMAN 15-111 & DoD Flips)	**1600** (AFMAN 15-111)
600 (DoD FLIPS)	**1 3/8** (DoD FLIPS)	**1000** (AFMAN 15-111)
500 (AFMAN 15-111 & DoD Flips)	**1 ¼** (DoD FLIPS)	**600** (AFMAN 15-111)
300 (AFMAN 15-111 & DoD Flips)	**1** (AFMAN 15-111 & DoD Flips)	
200 (AFMAN 15-111)	**¾** (AFMAN 15-111)	
100 (AFMAN 15-111)	**½** (AFMAN 15-111)	
	¼ (AFMAN 15-111)	

2.6.2. **Location.** The primary point of observation for Holloman Air Force Base is taken by the FMQ-19 which is located within the intersection of runways 16 and 22. The discontinuity sensor group at the approach end of runway 34 measures winds, visibility, and ceiling. Winds and visibility are also measured at the approach ends of runways 25 and the 04/07 intersection. An additional wind sensor takes measurements at the midfield intersection of runways 16/34 and 07/25.

2.6.3. **Shortfalls and Limiting Factors.** There are several limiting factors included in the following list. The weather technician's limited view of the horizon can prevent observation of some weather phenomena to the south and to the northeast due to hangars blocking the view. The automated Fixed Meteorological Equipment (FMQ-19) cannot determine a ceiling above 23,680 feet and cannot observe tornadoes, volcanic ash, snow depth, hail, ice pellets, funnel clouds, or smoke, limiting measurements during automated procedures. Finally, the high intensity lights northeast of building 577 and near the softball fields to the south may restrict the ability to determine sky condition and visibility during hours of darkness when augmentation procedures are being conducted.

2.6.4. **Augmentation.** When augmentation is required (supplementation or backup), the WF technician goes to the manual observation point approximately 30 feet east of the north corner of Building 577.

2.6.4.1. Mandatory supplementation criteria are found in AFMAN 15-111 and are implemented when the airfield is open.

2.6.4.2. In addition to the mandatory supplementation criteria, the weather technician will also supplement the observation when there are erroneous or non-reported ceilings below 3000 feet and visibility less than 3 miles, erroneous precipitation readings, when thunder is heard but not in the automated observation, and any time the automated observation is in error and could impact flight safety or operations.

2.6.5. **Dissemination of Observations.** The FMQ-19 automatically sends/disseminates weather observations through JET. When required, the weather technician will augment the observation in JET and then disseminate via JET.

2.6.6. **Backup Dissemination.** If JET is inoperative, observations will be disseminated verbally to ATC, RAPCON, SOF, and Airfield Management Operations. The observation will be disseminated long-line through AFW-WEBS, 25 OWS, another WF, or the National Weather Service (NWS).

2.7. Tropical Cyclone Products. The WF will use the Tropical Cyclone Threat Assessment Products produced by the responsible OWS (derived from National Hurricane Center or Joint Typhoon Warning Center) to provide WF customers forecasts of the expected onset, intensity, end times of significant winds, and closest point of approach for the associated tropical cyclone. No deviation from the official forecast position, track, movement, or intensity category is authorized.

2.8. Pilot-to-Metro Service (PMSV). During duty hours, the WF will provide full-service PMSV on the assigned frequency of 346.55 MHz or through ATC, RAPCON, or command post via phone patch. Aircrews are highly encouraged to relay pilot reports during PMSV, phone patches, or through the ATC/SOF. When the WF is not on duty, no PMSV service will be available.

Please see the closest PMSV location referenced in the US Instrument Flight Rating (IFR) Supplement during these times.

2.9. Space Weather Support. Space weather products will be provided, as requested, from the 2nd Weather Squadron via the Space Weather Products section located on AFW-WEBS.

2.10. Toxic Corridors. The WF will provide the 49 Civil Engineer Squadron Readiness Flight (49 CES/CEX) with weather information so they can calculate toxic corridors for chemical spills. The WF will not be responsible for producing toxic corridors. See section 8.9 of this publication for additional information.

2.11. Chemical Downwind and Effective Downwind Messages. The WF will provide Chemical Downwind and Effective Downwind Messages when requested by the 49 Civil Engineer Squadron Readiness Flight (49 CES/CEX) office for disaster response, chemical/nuclear attack, and exercise purposes.

2.12. Impacts to Operations. All weather impacts to operations are listed in **Attachment 3**.

Chapter 3

MISSION WEATHER ELEMENT

3.1. Mission Execution Forecast (MEF).

3.1.1. **Standard Day.** The Holloman MEF will be routinely posted at 0400, 1200, and 2000L when the airfield is open. If the airfield is closed during a MEF issuance time, that MEF will not be issued. MEFs will be valid until 2359L or until airfield closure, whichever is later.

3.1.2. **Non-standard Day.** The Holloman MEF will be posted three hours prior to the earliest takeoff (MQ-9s require issuance two hours prior to takeoff), unless coordinated the day prior, for each "go". The valid time will be from initial launch until airfield closure.

3.1.3. **Amendments.** This product will be amended as required to account for changing weather conditions that cross established forecast categories (see **paragraph 2.5.2** for specific references) or, at the forecaster's discretion, conditions that may adversely impact flying operations that are not represented in the product. The SOF will be notified by the ASE forecaster of the amendment. Flying unit Points of Contact (POCs), such as Top 3s and/or Directors of Operations (DOs), will be notified through their embedded weather forecaster (F-16s and MQ-9s) when an amendment is made to the MEF.

3.1.4. **Dissemination.** The MEF will be posted to the Base Weather SharePoint site under the link "Holloman MEF" located at: **https://usaf.dps.mil/sites/aetc-hmn/Weather/SitePages/Home.aspx**.

3.1.5. **Back Up Dissemination.** Backup method of delivery is via email. The second backup method of delivery is fax. Contact information is in **Table 3.1**.

Table 3.1. Flying Unit Points of Contact.

Unit	Phone Number (DSN)	Fax Number (DSN)	Organizational E-mail
6 ATKS	572-6973	572-0321	6RS/TOP3@us.af.mil
9/29 ATKS	572-5081	572-1216	29ATKSTop3@us.af.mil
311 FS	572-1171	572-2427	311FS_TOP3@us.af.mil
314 FS	572-4476	572-0740	314FS_TOP3@us.af.mil
8 FS	572-8881	None available	8FS.Top.3@us.af.mil
586 FLTS	572-1222	572-1023	None available
Army Air Division	349-1315	349-1694	usarmy.wsmr.atec.list.army-air-ops@mail.mil
Detachment 1, 82 Aerial Targets Squadron (ATRS)	572-5386	572-7978	None available

3.1.6. **MEF Product Description**

3.1.6.1. **Holloman MEF.** This link contains forecast weather data for Holloman AFB and applicable local ranges for the local flying day (through 2359L or airfield closure, whichever is later).

3.1.6.2. **Holloman Hazards.** This link contains flying hazards including turbulence, icing, and thunderstorms for the flying ranges and the local flying area. They will only be produced for the current day's flying window.

3.1.6.3. **Flight Level Winds.** This link contains flight-level winds data for all of the flying ranges in 6 hour blocks.

3.1.6.4. **Absolute Humidity.** This link contains flight level absolute humidity data for all of the flying ranges in 6 hour blocks.

3.1.6.5. **MOA Flare Restrictions.** This link contains the flare restrictions for the Holloman AFB flying areas. The information contained on this slide is generated by the US Forest Service Wildland Fire Assessment System (USFS-WFAS). USFS-WFAS is responsible for the content of this slide.

3.1.6.6. **Ranges Fire Danger.** This link contains the current fire danger restriction for each Holloman AFB range as determined by the Range Management Office. Information about current National Weather Service Red Flag Warnings are also included here.

3.1.6.7. **Holloman 5-Day Forecast.** This link contains the Holloman AFB 5-day forecast

3.1.6.8. **Solar/Lunar/Illumination.** This link contains solar/lunar data as well as illumination data broken into tabs for each month of the year.

3.2. MISSIONWATCH.

3.2.1. The WF will conduct a continuous MISSIONWATCH of all routes and flying areas used by Holloman flying units. The WF will provide the SOF and flying units updates for weather affecting airfield operations and area missions via the MEF, phone notification, or through their embedded forecaster. MISSIONWATCH criteria will be based off of aircraft sensitivities and mission requirements. If weather conditions change and a MEF amendment is needed (see **paragraph 2.5.2** for criteria), the WF will amend and reissue the MEF with updated current and forecast conditions.

3.2.2. **Flying Unit Points of Contact.** All flying units can be contacted for any significant changes in the observation or amendments to the forecast as shown in **Table 3.1**.

3.2.3. **Impacts to Operations.** All weather impacts to operations are listed in **Attachment 3**.

3.3. Flight Weather Brief. The WF will provide flight weather briefings using DD Form 175-1 or verbal briefings based on the pilot's requirement for all assigned aircraft, transient aircraft (during duty hours if time permits), and aircraft contacted through PMSV. If duty priorities preclude timely completion of the briefing, weather technicians will direct transient aircrews to 25 OWS or their home-station WF.

3.4. Squadron Integration.

3.4.1. **MQ-9s** . Supported MQ-9 units include 6th Attack Squadron (6 ATKS), 9th Attack Squadron (9 ATKS), and 29th Attack Squadron Support (29 ATKS). The WF will assign

weather technicians to provide weather support to include the production of the MEF, Range WWAs, weather briefings, and MISSIONWATCH. This forecaster will be co-located in the operations center at building 302 (6 ATKS squadron building) during airfield operating hours until September, 2020. After September, 2020, they will be located in the operations center in building 318 (9/29 ATKS squadron building). Outside of flying hours or at WF leadership's direction, this forecaster will be assigned to building 577.

3.4.2. **F-16s.** Supported F-16 units include 8 Fighter Squadron (8 FS), 311 Fighter Squadron (311 FS) and 314 Fighter Squadron (314 FS). The WF will assign weather forecasters to provide tailored weather products and briefings to aircrews in support of fighter squadron training syllabus objectives. The 311 FS and 314 FS forecasters will be located at building 892 in the training and evaluations section during scheduled F-16 flying. The 8 FS forecaster will be located at the operations (Top 3) desk at building 1062. Outside of flying hours or at WF leadership's direction, these forecasters will be assigned to building 577.

Chapter 4

STAFF WEATHER ELEMENT

4.1. Wing Staff Meetings. The content of the briefing will consist of one briefing slide containing a 5-day outlook with an operational impacts stoplight chart. This product will be briefed in person to 49 WG leadership as required.

4.2. Installation Command Center/Crisis Action Team. The format of the briefing will be tailored to the scenario driving the meeting. The briefing may include the current weather, 5-day outlook for Holloman AFB, enroute weather and hazards, and forecast destination weather.

4.3. Deployment Concept Briefing. The focus of this briefing will be on departure conditions and deployed area conditions and climatology. As a minimum, the briefing will include forecast conditions at Holloman AFB for departure time, deployed area climatological conditions, and forecast conditions for destination arrival time. Other topics can be briefed upon request.

4.4. Instrument Refresher Course (IRC). A WF representative will provide a local weather effects briefing to all IRC classes for host and tenant units as manning permits. IRC schedulers will provide the WF with a schedule of upcoming IRCs as early in the process as possible.

4.5. Climatology Support. Climatology data will be available upon request. This includes requests for data for other locations. Requests will be made at least five duty days in advance in order to allow the WF time to research and compile the data.

4.6. Air Traffic Control Training/Orientation. The WF will provide orientation, training, and limited observing certification for ATC personnel. Training/certification can be conducted Monday through Friday between the hours of 0800 and 1600L. ATC personnel will contact the WF to coordinate a training time.

4.7. Non-Standard Requests. WF personnel will support official meteorological request as duty priorities permit. Requests from outside the DoD will be coordinated through Public Affairs.

Chapter 5

WATCHES, WARNINGS, AND ADVISORIES

5.1. General.

5.1.1. The 25 OWS and the WF will jointly monitor observed and forecast weather conditions to notify selected agencies when pre-established weather condition criteria occur or are expected to occur.

5.1.2. **Desired Lead Time (DLT).** Advance warning of threatening weather conditions allows local agencies to take specific actions prior to occurrence. The desired lead time is the minimum amount of advanced notice an agency requires prior to the onset of a particular weather phenomenon.

5.1.3. All WWAs are issued in accordance with (IAW) AFMAN 15-129. Warnings and advisories may be issued as forecast or observed.

5.2. Agency Responsibilities.

5.2.1. The 25 OWS will:

5.2.1.1. In the event of unforeseen/emergency situations, issue required WWAs in a timely manner to meet full desired lead time or when the weather event is imminent. This includes complete communications outages and AOL evacuations while in transit.

5.2.2. The WF will:

5.2.2.1. Issue all forecast and observed weather watches, warnings, and advisories for Holloman AFB, Centennial Range, Oscura Range, and Red Rio Range as required.

5.2.2.2. Disseminate weather WWAs via JET.

5.2.2.3. Maintain a full WWA dissemination listing in accordance with AFMAN 15-129. This document will be maintained separately from this instruction for ease of use and update.

5.2.3. The 49 WG Command Post will:

5.2.3.1. Monitor weather information disseminated through JET.

5.2.3.2. Acknowledge receipt of automated JET WWA notifications.

5.2.3.3. Disseminate weather WWAs to required agencies in accordance with locally-maintained checklists.

5.3. Weather Watches. The WF will issue a forecast weather watch for within 5 NM of the center of the Holloman AFB runway complex when the potential exists for the criteria defined in **Table 5.1** Severe weather action plan criteria is defined in **chapter 6.**

Table 5.1. Forecast Weather Watch Criteria and Desired Lead Times.

		Criteria	Desired Lead Time
Severe Weather Action Plan Criteria	1	Tornado	As potential warrants
	2	Damaging Winds ≥ 50 knots	
	3	Freezing Precipitation	
	4	Severe Thunderstorms (Damaging hail ≥ 3/4 inch and/or damaging winds ≥ 50 knots)	
Non-severe forecast watch criteria	9	Lightning within 5 NM	30 minutes

5.4. Weather Warnings.

5.4.1. **Forecast Weather Warnings** . The WF will issue a forecast weather warning for within 5 NM of the center of the Holloman AFB runway complex when the criteria defined in **Table 5.2** is occurring or is expected to occur.

Table 5.2. Forecast Weather Warning Criteria and Desired Lead Times.

		Criteria	Desired Lead Time
Severe Weather Action Plan Criteria	1	Tornado	15 minutes
	2	Damaging Winds ≥ 50 knots	60 minutes
	3	Freezing Precipitation	
	4	Severe Thunderstorms (Damaging hail ≥ 3/4 inch and/or damaging winds ≥ 50 knots)	
Non-severe forecast warning criteria	5	Blizzard Conditions (Winds ≥ 30 knots and Visibility ≤ 1/4SM in Snow/Blowing Snow)	
	6	Heavy Rain (≥ 2 inches in 12 hours)	
	7	Heavy Snow (≥ 2 inches in 12 hours)	
	8	Moderate Thunderstorms (Large hail ≥ 1/4 inch but < 3/4 inch and/or strong winds 35-49 knots)	
	9	Strong Winds ≥ 35 but < 50 knots	

5.4.2. **Observed Weather Warnings** . The WF will issue an observed weather warning for events within 5 NM of Holloman AFB runway complex or within 10 NM of either the north or south end of the High Speed Test Track (HSTT) when the criteria defined in **Table 5.3** occurs.

Table 5.3. Observed Weather Warning Criteria and Desired Lead Times.

	Criteria	Desired Lead Time
1	Lightning (Within 10 NM of the Test Track)	Observed
2	Lightning (Within 5 NM)	

5.5. Weather Advisories.

5.5.1. **Forecast Weather Advisories.** The WF will issue a forecast weather advisory for within 5 NM of the center of the Holloman AFB runway complex when the criteria defined in **Table 5.4** is occurring or is expected to occur.

Table 5.4. Forecasted Weather Advisory Criteria and Desired Lead Times.

	Criteria	Desired Lead Time
1	Surface winds ≥ 25 but < 34 knots	30 minutes

5.5.2. **Observed Weather Advisories.** The WF will issue an observed weather advisory when the airfield is open when the criteria defined in **Table 5.5** occurs. Advisories #2 through #4 and #6 through #14 in **Table 5.5** are only issued when the airfield is open.

Table 5.5. Observed Weather Advisory Criteria and Desired Lead Times.

	Criteria	Desired Lead Time
1	Observed Blowing Dust with Visibility ≤ 5 statute miles	Observed
2	Surface temperature > 32ºF but ≤ 40ºF	
3	Surface temperature ≤ 32ºF	
4	Thunderstorms within 25 miles of Holloman AFB	
5	Thunderstorms within 10 miles of Holloman AFB	
6	Ice Foreign Object Debris (temp ≥ -4ºF and < 50ºF and dew point within 5ºF of the temp)	
7	Surface winds sustained ≥ 35 kts occurring in the Local Flying Area (LFA)	
8	Light or greater icing in the LFA	
9	Moderate or greater turbulence in the LFA	
10	Low-Level Wind Shear (LLWS) on station	
11	Observed Crosswinds 15-19 knots	
12	Observed Crosswinds 20-24 knots	
13	Observed Crosswinds ≥ 25 knots	
14	Observed Surface Winds ≥ 31 knots	

5.6. Watches, Warnings, and Advisories for Local Ranges. The WF will issue a weather watch or warning for Oscura, Red Rio, and/or Centennial range whenever the airfield is open when the criteria in **Table 5.6** occurs. These Watches and Warnings will not be issued on weekends and holidays unless previously coordinated with Range Management. The required watches and warnings are the same for all three ranges.

Table 5.6. Weather Watch and Warning Criteria and Desired Lead Times.

	Criteria	Desired Lead Time
Watches	Tornado	As potential warrants
	Damaging Winds ≥ 45 knots	As potential warrants
	Severe thunderstorm (hail ≥ ¾" diameter and/or winds ≥ 45 kts)	As potential warrants
	Lightning within 10 NM	30 minutes
Warnings	Tornado	15 minutes
	Damaging Winds ≥ 45 knots	60 minutes
	Severe thunderstorm (hail ≥ ¾" diameter and/or winds ≥ 45 kts)	60 minutes
	Lightning within 10 NM	Observed

5.7. Dissemination of Watches/Warnings/Advisories. All WWAs will be disseminated using JET.

5.7.1. All WWAs will be disseminated using JET.

5.7.2. If JET is inoperative:

5.7.2.1. During WF duty hours, the WF will contact Command Post, Tower, and Airfield Management to disseminate WWAs verbally.

5.7.2.2. During non-WF hours, the WF's standby forecaster will be responsible for disseminating the WWAs to Holloman AFB via back-up methods.

5.8. WWA Text Format. The text of WWAs will contain the corresponding number, the specific valid time period (or "until further notice" for observed criteria), and specific conditions expected. They will be numbered consecutively with the number of the month followed by the number of the WWA for that month.

Chapter 6

SEVERE WEATHER ACTION PLAN

6.1. General. These procedures are in place to ensure sufficient personnel are available during potential or actual severe weather events. For the purposes of these procedures, severe weather is defined as any weather considered critical enough by Holloman AFB to require advance/special notice and subsequent actions to prevent serious injury or damage to personnel, property, or resources.

6.2. Responsibilities. The WF will perform the SWAP as defined in AFMAN 15-129, AFMAN 10-206, *Operational Reporting*, and this document. More specifically, the WF will accomplish the following procedures:

6.2.1. **Notification.** The on WF duty weather technician or the Command Post Controller will notify the SWMT standby member(s) according to the following guidance:

6.2.1.1. During WF duty hours, the weather technician will implement SWAP by notifying the SWMT whenever any WWAs in **Table 6.1** are issued.

Table 6.1. Weather Watches and Warnings Requiring SWMT Leader Notification.

#	Criteria
1	Tornado
2	Surface winds ≥ 50 kts
3	Severe Thunderstorms
4	Freezing Precipitation

6.2.2. **Activation.** The WF will be on duty in the weather station when a weather watch or warning from **Table 6.1** is issued. During WF non-duty hours, this will require the standby weather technician to report to duty after they have notified the SWMT leader. The SWMT leader may (in the case of a warning, will) brief the applicable watches and/or warnings and other pertinent weather information to 54 OSS leaders, 54 FG staff, 49 OG staff, 49 MXG operations center, 49 MSG staff, and 49 WG leadership through an established Primary, Alternate, Contingency, and Emergency (PACE) plan for each agency. Updates will be provided to these same personnel as requested or warranted. The SWMT leader will recall additional weather technicians (or place on standby) and execute SWAP duties/responsibilities as the situation warrants.

6.2.3. **OPREP-3 Procedures.** When severe weather causes damage or injury at Holloman AFB, the WF or 25 OWS may be asked to provide information to the Command Post in accordance with AFI 10-206. This information will be provided in a time-sensitive manner.

Chapter 7

MISSION AND REQUIREMENTS OF SUPPORTED AGENCIES

7.1. General. The WF supports numerous units on Holloman AFB, each with their own unique mission. Weather information is also briefed to enhance resource protection for the base populace as well as those conducting range operations locally. Unmanned Airfield Operations are not provided WF support outside airfield operating hours with the exception of WF response to SWAP events.

7.2. 49th Operations Group (6 ATKS, 9 ATKS and 29 ATKS).

7.2.1. **Aircraft/Mission.** MQ-9: Pilot and sensor operator training.

7.2.2. **Weather Support Requirements.** The WF will provide MEFs and MISSIONWATCH for local flying operations and missions in scheduled flying areas.

7.3. 54th Fighter Group (8 FS, 311 FS and 314 FS)

7.3.1. **Aircraft/Mission.** Basic Course F-16 pilot training.

7.3.2. **Weather Support Requirements.** The WF will provide a to-the-point tailored weather brief based off of the Holloman and Range MEFs. Additionally, they will provide embedded weather support to mass and step briefings before each departure. Finally, MISSIONWATCH will be provided for these missions.

7.4. 586th Flight Test Squadron (586 FLTS).

7.4.1. **Aircraft/Mission.** T-38B, C-12, and other assigned aircraft tasked to support their mission depending on specific test requirements. Evaluates systems and subsystems for aircraft, missiles, and space vehicles.

7.4.2. **Weather Support Requirements.** The WF will provide MEFs and MISSIONWATCH for local flying operations and missions in scheduled Flying Areas.

7.5. 82nd Aerial Targets Squadron.

7.5.1. **Aircraft/Mission.** QF-16: DoD research, development, and test projects. Supervises and monitors the operations and maintenance of QF-16 drones.

7.5.2. **Weather Support Requirements.** The WF will provide MEFs and MISSIONWATCH for local flying operations and missions in scheduled Flying Areas.

7.6. Aero Club.

7.6.1. **Aircraft/Mission.** Diamond Aircraft DA-40, T-41, and Cessna 210: Provides Federal Aviation Administration certified **part 141** flying courses for aircraft-type rating, Private Pilot Instrument, Commercial, and Airline Transport Pilot certification.

7.6.2. **Weather Support Requirements.** The WF provides planning weather on the SharePoint page, as well as resource protection WWAs. The WF can only provide flight weather briefings to Aero Club members performing official Air Force operational duties.

7.7. Acoustic Research Complex (ARC).

7.7.1. **Aircraft/Mission** . Air Force Research Laboratory (AFRL) personnel measure acoustic signatures of aircraft flown through the complex at various engine/power settings and configurations.

7.7.2. **Weather Support Requirements** . The WF will provide general planning weather, MEFs and MISSIONWATCH for local flying operations and execution forecasts when ARC personnel are not available.

7.8. Army Air.

7.8.1. **Aircraft/Mission.** C-12, UH-60: Provide support to the White Sands Missile Range through activities including aerial photography, missile recovery, cargo and personnel transportation, and experimental missions.

7.8.2. **Weather Support Requirements.** The WF will provide verbal weather briefings or MEFs and MISSIONWATCH for local flying operations and missions in scheduled flying areas.

Chapter 8

RECIPROCAL SUPPORT

8.1. General. This chapter outlines WF and individual unit responsibilities.

8.2. 49th Wing.

8.2.1. 49th Wing personnel will:

8.2.1.1. Coordinate and schedule with the WF all upcoming exercise support.

8.2.1.2. Advise the WF of any changes/updates to all base Host/Tenant Unit Agreements.

8.2.2. The WF will:

8.2.2.1. Manage WF personnel and resources to support base operations/exercises.

8.2.2.2. Provide staff weather support for routine briefings and inclement weather notifications.

8.3. Manned Flying Units.

8.3.1. Flying unit personnel will:

8.3.1.1. Notify weather personnel of flight scheduling changes no later than 12 hours in advance or as soon as possible to ensure forecaster personnel are available and scheduled to support as required (i.e., brief and take off times, route, range, or flight cancellations).

8.3.1.2. Ensure the weather technician receives information required to produce Target Acquisition Weather Software (TAWS) products. (At this time, the TAWS software is inoperative due to an Air Force-wide compatibility issue.)

8.3.1.3. Fill out electro-optical feedback worksheets for all missions for which they need to receive TAWS.

8.3.1.4. Pass all significant PIREPs to the WF through PMSV, or phone patch through RAPCON or ATC Tower.

8.3.1.5. Inform the WF of weather communications equipment outages when necessary. Coordinate alternative method of dissemination of weather information.

8.3.1.6. Provide feedback on MEF products and tailored weather briefings to allow for verification and improve product performance.

8.3.1.7. **(311/314/8 FS only)** Provide an appropriate workspace for the embedded weather forecaster to accomplish his/her assigned duties that is within acceptable distance of the Top 3 desk.

8.3.1.8. Utilize their assigned embedded weather forecaster assigned to the F-16 fighter squadrons for all mission weather support during their flying windows. When a forecaster is actively working over at building 892 and/or 1062, the airfield services element at building 577 is authorized to redirect all fighter squadron Requests for Information (RFI) to the embedded weather forecaster. Concerns with the assigned weather forecaster will be forwarded to the weather flight chief or commander.

8.3.2. The WF will:

8.3.2.1. Ensure tailored briefing products are disseminated properly and in a timely manner to supported unit Top 3s and mission planners. They will troubleshoot if not disseminated correctly.

8.3.2.2. Debrief pilots upon termination of flight, for any differences between actual and briefed weather.

8.3.2.3. When notified by the flying unit of problems with JET, assist in troubleshooting the problem and provide an alternate means of disseminating weather information.

8.3.2.4. Provide at least 12 hours' notice to supported unit DOs and Top 3s of the need to recall the embedded weather forecaster back to the WF to support mission requirements.

8.4. Remotely Piloted Aircraft (RPA) Units.

8.4.1. Flying unit personnel will:

8.4.1.1. Accomplish all tasks listed in paragraphs **8.3.1.1**, **8.3.1.2**, and **8.3.1.4** through **8.3.1.6**.

8.4.1.2. Provide orientation and indoctrination training to new weather personnel by appointment.

8.4.1.3. Provide a workspace and equipment for the embedded weather forecaster.

8.4.1.4. Provide a workspace and equipment for the weather flight's AOL.

8.4.1.5. Utilize their assigned embedded weather forecaster assigned to the MQ-9 operations center for all mission weather support during their flying windows. When a forecaster is actively working over at the designated MQ-9 operations center, the airfield services element at building 577 is authorized to redirect all MQ-9 weather-related RFIs to the embedded weather forecaster. Concerns with the assigned weather forecaster will be forwarded to the WF chief or commander.

8.4.2. The WF will:

8.4.2.1. Accomplish all tasks listed in Section 8.3.2.

8.4.2.2. When integrated, provide in-person weather support to the Top 3s at the operations desk throughout the flying day.

8.5. 49th Wing Command Post (49 WG/CP).

8.5.1. 49 WG/CP will:

8.5.1.1. Disseminate WWAs.

8.5.1.2. Notify the WF of weather communications equipment outages.

8.5.1.3. Channel to higher command, via operational reporting, actual occurrences of severe weather as required. 49 WG/CP will coordinate draft report with the WF or 25 OWS prior to posting the report.

8.5.1.4. Provide orientation and indoctrination training to new weather personnel by appointment.

8.5.2. The WF will:

8.5.2.1. Ensure all WWA notifications take place via JET or backup method.

8.5.2.2. When notified by 49 WG/CP of problems with JET, assist in troubleshooting the problem and provide an alternate means of disseminating weather information if the equipment must be logged out.

8.5.2.3. Provide information, as requested, to be included in OPREP-3 reports.

8.5.2.4. Provide the Command Post with an updated memorandum with the standby weather phone contact information as well as current contact information for flight leadership. WF updates to this memorandum will be provided as required.

8.6. 49th Wing Safety (49 WG/SE).

8.6.1. 49 WG/SE will notify the WF as soon as possible of all known aircraft mishaps or incidents requiring weather information.

8.6.2. The WF or OWS will:

8.6.2.1. Provide all available weather information related to aircraft accident or mishap investigations.

8.6.2.2. Provide a qualified weather representative for safety investigation boards, as requested.

8.6.2.3. Provide seasonal briefings on weather-related aviation hazards at quarterly aircrew Flight Safety Meetings upon request.

8.7. 49th Wing Public Affairs (49 WG/PA).

8.7.1. 49 WG/PA will:

8.7.1.1. Notify base population via marquee of severe weather information when deemed necessary by the wing commander.

8.7.1.2. Approve release of weather information to non-DoD agencies and the public based on public release regulations and consultation on legality of providing the requested data.

8.8. 49th Civil Engineer Squadron.

8.8.1. 49th Civil Engineer Squadron Operations (49 CES/CEO) will:

8.8.1.1. Provide emergency backup power for WF operations.

8.8.1.2. Notify the WF 3 days prior to any work on Building 577's power supply and/or generators. The WF reserves the right to delay maintenance, testing, or modification to Building 577's power supply, commercial or generator, during operating hours when such maintenance could degrade MISSIONWATCH support to local flying or METWATCH support when base resource protection is an issue.

8.8.2. The WF will:

8.8.2.1. Provide local climatology summary to 49 CES as required.

8.9. 49th Civil Engineer Squadron Readiness Flight (49 CES/CEX).

8.9.1. 49 CES/CEX will:

8.9.1.1. Calculate toxic hazard corridors for chemical spills. CEX will use onsite weather data in conjunction with indigenous weather information provided by the WF.

8.9.2. The WF will:

8.9.2.1. Provide chemical and effective downwind messages to the Nuclear, Biological, and Chemical (NBC) Cell upon request.

8.9.2.2. Provide weather data required for chemical spills, emergencies, or disasters.

8.9.2.3. Partner with the Civil Engineer Squadron Readiness Flight (CEX), Fire Emergency Services (CEF), installation Medical Group's Bioenvironmental Engineer Flight, Army Installation Directors of Emergency Services, and National Guard Civil Support Teams for Air National Guard (ANG) weather organizations, as the Weather SME responsible for optimizing weather data input to Chemical Downwind Messages (CDMs), Effective Downwind Messages (EDMs), and Chemical, Biological, Radiation, and Nuclear (CBRN) hazard-prediction models used by these Emergency Support Functions (ESFs) for decision assistance in the Emergency Operations Center (EOC), CBRN Control Center, and at the incident site.

8.9.2.4. Advise and provide the most accurate and representative observed and/or forecast alphanumeric and gridded meteorological data type appropriate to a particular CBRN event to users employing Areal Locations of Hazardous Atmospheres (ALOHA), Computer-Aided Management of Emergency Operations (CAMEO), and CBRN hazard-prediction (i.e., "plume") models resident in the Joint Warning and Reporting Network (JWARN)/Joint Effects Model (JEM)/Joint Operational Effects Federation (JOEF) architecture IAW AFI 10-2501, AFMAN 10-2503 and equivalent Joint guidance, to ensure consistency between CBRN hazard area predictions and the installation forecast.

8.10. 49th Communications Squadron (49 CS).

8.10.1. 49 CS will:

8.10.1.1. Provide routine and emergency maintenance for communications and computer equipment located on Holloman AFB not covered by contractor logistics. This includes standard maintenance to the JET server located in Building 202 in accordance with the Service-Level Agreement (SLA) on file.

8.10.2. The WF will:

8.10.2.1. Notify 49 CS of network or system outages in accordance with the SLA on file.

8.11. 54th Operations Support Squadron Radar, Weather and RF Transmission Systems (RAWS) Flight (54 OSS/OSAM).

8.11.1. RAWS will:

8.11.1.1. Provide routine and emergency maintenance for weather observing and weather-related communications equipment located at or near Holloman AFB not covered by contractor logistics. Reference TO 31M1-2FMQ19-1 for frequency and procedures for conducting maintenance.

8.11.1.2. Follow the priorities established IAW 54th Fighter Group, 49th Mission Support Group, and White Sands Missile Range Letter of Agreement guidance for restoring inoperative weather sensing and communication equipment. Maintenance personnel will provide 2 days' notice to the WF before taking equipment down for maintenance. RAWS will also coordinate preventive maintenance downtimes with the WF.

8.12. 54th Operations Support Squadron Airfield Management Operations (54 OSS/OSAA).

8.12.1. 54 OSS/OSAA will:

8.12.1.1. Disseminate weather advisories and warnings in accordance with local procedures.

8.12.1.2. Notify the WF of weather communications equipment outages.

8.12.1.3. Provide the WF with changes in the runway condition.

8.12.1.4. Provide airfield orientation to new weather personnel by appointment.

8.12.1.5. When requested by the WF, submit weather information changes to FLIPS to the Federal Aviation Administration.

8.12.1.6. When requested by the WF, issue Notices to Airmen (NOTAMS) of weather equipment outages and limitations.

8.12.2. The WF will:

8.12.2.1. Ensure all WWA notifications via JET or backup methods as outlined in this instruction.

8.12.2.2. Notify Airfield Management of all PMSV radio outages and restorations to service.

8.12.2.3. When notified by 54 OSS/OSAA of problems with JET, assist in troubleshooting the problem and provide an alternate means of disseminating weather information, if the equipment must be logged out.

8.13. 54th Operations Support Squadron Tower Section (54 OSS/OSAT).

8.13.1. 54 OSS/OSAT will:

8.13.1.1. Participate in the CWW IAW AFMAN 15-111, section 4.9.

8.13.1.2. Notify the WF of weather communications equipment outages.

8.13.1.3. Notify the WF of changes in the active runway.

8.13.1.4. Relay all PIREPs to the WF within 10 minutes of receipt.

8.13.1.5. Provide tower orientation to new weather personnel by appointment.

8.13.1.6. Report tower visibility different from the prevailing surface visibility if either is less than 4 statute miles.

8.13.1.7. Report occurrence of previously unreported weather conditions that could affect flight safety or be critical to the safety or efficiency of other local operations and resources.

8.13.2. The WF will:

8.13.2.1. Ensure all WWA notifications via JET or backup methods as outlined in this instruction.

8.13.2.2. Incorporate ATC tower-reported weather observations provided under the CWW concept into weather observations.

8.13.2.3. Notify ATC Tower of all PMSV radio outages and restorations to service.

8.13.2.4. Notify ATC Tower when evacuating the WF facility.

8.13.2.5. Provide weather orientation, training, and certification briefings for ATC Tower controllers by appointment.

8.13.2.6. When notified by 54 OSS/OSAT of problems with JET, assist in troubleshooting the problem and provide an alternate means of disseminating weather information, if the equipment must be logged out.

8.14. 54th Operations Support Squadron Radar Approach Control Section (54 OSS/OSAR).

8.14.1. 54 OSS/OSAR will:

8.14.1.1. Relay all weather-related PIREPs to the WF.

8.14.1.2. Notify the WF of weather communications equipment outages.

8.14.1.3. Provide orientation to new weather personnel by appointment when requested.

8.14.2. The WF will:

8.14.2.1. Ensure all WWA notifications via JET or backup method.

8.14.2.2. Notify RAPCON of all PMSV radio outages and restorations to service.

8.14.2.3. Notify RAPCON when evacuating the WF facility.

8.14.2.4. Provide weather orientation briefings for RAPCON controllers by appointment.

8.14.2.5. When notified by 54 OSS/OSAR of problems with JET, assist in troubleshooting the problem and provide an alternate means of disseminating weather information, if the equipment must be logged out.

8.15. 54th Operations Support Squadron Weapons and Tactics Flight (54 OSS/OSK).

8.15.1. 54 OSS/OSK will:

8.15.1.1. Notify the WF of the date, time, mission commander, and other pertinent information for verification planning meetings.

8.15.1.2. Assist the WF by providing training to WF members covering weather-related impacts to weapons and tactics.

8.15.2. The WF will:

8.15.2.1. Provide briefings on WF services, flight weather hazards, weather codes, and charts at requested briefings.

8.16. 54th Operations Support Squadron Operations Flight (54 OSS/OSO).

8.16.1. 54 OSS/OSO will:

8.16.1.1. Provide information relating to the planned missions of 49 WG and tenant unit missions that utilize 49 WG airspace.

8.16.1.2. Provide information related to air refueling tracks utilized by aircraft assigned to or performing temporary operations from Holloman AFB.

8.16.1.3. Provide monthly metrics pertaining to sorties generated by 49 WG aircraft.

8.16.2. The WF will:

8.16.2.1. Provide weather data and planning information as requested.

8.17. 846th Test Squadron (846 TS).

8.17.1. 846 TS will:

8.17.1.1. Request support for test track missions during 846th TS Staff Meteorologist's TDY/leave no less than one month in advance.

8.17.1.2. Provide the WF with a copy of the test track schedule.

8.17.1.3. Provide the WF with format and specific instructions for mission support and reports.

8.17.1.4. Provide test date, times, and specific requirements for each mission.

8.17.1.5. Provide copies of operations plans, weather annexes, and other information to the WF.

8.17.1.6. Provide orientation and indoctrination training to new weather personnel by appointment.

8.17.2. The WF will:

8.17.2.1. If manpower levels allow, provide backup support for 846 TS test track missions when the staff meteorologist is on leave. If necessary, obtain training from the staff meteorologist.

8.17.2.2. Request mission weather go/no go criteria from the staff meteorologist prior to test readiness review. If required, the weather technician will advise test manager on weather go/no go criteria.

8.17.2.3. If operational requirements permit, provide one weather technician for the operational test phase and L-1 day test readiness review meeting.

8.17.2.4. Prepare 24-hour weather report after the mission.

8.18. Acoustic Research Complex (ARC) Program (AFLCMC/XZIG) .

8.18.1. AFRL ARC Program will:

8.18.1.1. Provide a schedule of test days and back-up test days, times of testing, and expected set-up/tear down schedule at least 2 weeks prior (provide more notice if possible).

8.18.1.2. Provide aircraft type and owning organization (with POCs), if possible.

8.18.2. AFLCMC/XZIG will:

8.18.2.1. Provide staff meteorological support to the Program Test Director (PTD) for ARC tests.

8.18.2.2. Provide the consensus, official weather forecast for tests and set-up/tear down for tests for the ARC to the PTD.

8.18.2.3. Arrange for alternate weather support when WF or AFLCMC/XZIG are not able to provide support.

8.18.2.4. Coordinate with 557th Weather Wing and other government weather organizations.

8.19.1. The WF will:

8.19.1.1. Provide general planning weather, MEFs and MISSIONWATCH for local flying operations area, briefings, and execution forecasts when AFLCMC/XZIG is not available.

8.19.1.2. Provide METWATCH of ARC site during test times and during set-up/tear down times through AFLCMC/XZIG or as directed by the PTD for the ARC, (as resources and schedule permit).

8.20. 49th Security Forces Squadron (49 SFS).

8.20.1. 49 SFS will:

8.20.1.1. Security Forces will notify the WF of observed hail, tornadoes, freezing precipitation, or other significant weather encountered during routine patrols during WF duty hours.

8.20.2. The WF will:

8.20.2.1. Provide requested weather information for Security Forces operations.

8.20.2.2. Provide weather observer training, as required.

8.21. 49th Operational Medical Readiness Squadron, Bioenvironmental Flight (49 OMRS/SGPB).

8.21.1. 49 OMRS/SGPB will:

8.21.1.1. IAW AFI 48-151, issue and disseminate installation heat stress flag conditions and Fighter Index of Thermal Stress (FITS) conditions.

8.21.2. The WF will:

8.21.2.1. Provide atmospheric temperature information through JET to calculate FITS condition.

8.22. NWS El Paso Field Office.

8.22.1. The NWS El Paso Field Office will:

8.22.1.1. Adjust the Holloman weather radar's Volume Coverage Pattern (VCP) when requested by a forecaster from the WF.

8.22.1.2. Coordinate with the WF on biannual Unit Radar Committee (URC) meetings.

8.22.1.3. Advise the WF's radar managers on local policy pertaining to VCP settings.

8.22.2. The WF will:

8.22.2.1. Call the National Weather Service's radar technician when weather conditions drive the need to change the Tularosa radar's VCP.

8.22.2.2. Relay pertinent information concerning future upgrades/construction briefed at URC meetings to 54 OSS/DO and 54 OSS/CC for their situational awareness.

8.22.2.3. Bring outstanding issues with the radar support to the attention of NWS personnel as required.

RYAN P. KEENEY, Colonel, USAF
Commander, 49th Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 10-2501, *Emergency Management Program,* 9 March 2020

AFI 15-128, *Weather Force Structure,* 20 June 2019

AFI 48-151, *Thermal Injury Prevention Program,* 21 April 2020

AFMAN 10-206, *Operational Reporting (OPREP),* 31 August 2020

AFMAN 10-206 AETC Supplement, *Operational Reporting (OPREP),* 27 November 2018

AFMAN 11-202v3, *Flight Operations,* 9 June 2020

AFMAN 15-111, *Surface Weather Observations,* 11 March 2019

AFMAN 15-124, *Meteorological Codes,* 15 January 2019

AFMAN 15-129, *Air and Space Weather Operations,* 8 July 2020

AFMAN 91-203, *Air Force Occupational Safety, Fire and Health Standards,* 10 December 2018

DESR 6055.09_AFMAN 91-201, *Explosives Safety Standards,* 27 May 2020

AFPD 15-1, *Weather Operations,* 13 November 2019

Prescribed Forms

None

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

DD Form 175-1, *Flight Weather Briefing*

Abbreviations and Acronyms

6 ATKS—6th Attack Squadron

9 ATKS—9th Attack Squadron

25 OWS—25th Operational Weather Squadron

29 ATKS—29th Attack Squadron

49 CES—49th Civil Engineer Squadron

49 CES/CEA—49th Civil Engineer Squadron Environmental Flight

49 CES/CEF—49th Civil Engineer Squadron Fire Department

49 CES/CEO—49th Civil Engineer Squadron Operations Flight

49 CES/CEP—49th Civil Engineer Squadron Programs Flight

49 CES/CEX—49th Civil Engineer Squadron Readiness Flight

49 CS—49th Communication Squadron

49 MDG—49th Medical Group

49 MXG/CC—49th Maintenance Group Commander

49 MXS/MXM—49th Maintenance Squadron Production Section

49 MXS/MXMW—49th Maintenance Squadron Munitions Flight

49 OG—49th Operations Group

49 OG/CC—49th OG Commander

49 OMRS/SGPB—49th Operational Medical Readiness Squadron Bioenvironmental Flight

49 WG/CP—49th Wing Command Post

54 OSS/CC—54th Operations Support Squadron Commander

54 OSS/OSAA—54th Operations Support Squadron Airfield Management Flight

54 OSS/OSAM—54th Operations Support Squadron METNAV Flight

54 OSS/OSAR—54th Operations Support Squadron RAPCON Flight

54 OSS/OSAT—54th Operations Support Squadron Air Traffic Control Flight

54 OSS/OSK—54th Operations Support Squadron Weapons and Tactics Flight

54 OSS/OSO—54th Operations Support Squadron Operations Flight

54 OSS/OSW—54th Operations Support Squadron Weather Flight

49 SFS—49th Security Forces Squadron

49 WG—49th Wing

49 WG/PA—49th Wing Public Affairs

49 WG/SE—49th Wing Safety

82 ATRS—82nd Aerial Targets Squadron

311 FS—311th Fighter Squadron

314 FS—314th Fighter Squadron

54 FG/CC—54th Fighter Group Commander

586 FLTS—586th Flight Test Squadron

704 TG/CC—704th Test Group Commander

846 TS—846th Test Squadron

AFB—Air Force Base

AFI—Air Force Instruction

AFLCMC/XZIG—Air Force Research Laboratory Acoustic Research Complex Program

AFMAN—Air Force Manual

AFPD—Air Force Policy Directive

AFRL—Air Force Research Laboratory

AFRIMS—Air Force Records Information Management System

AFW-WEBS—Air Force Weather Web

AGE—Aerospace Ground Equipment

AIREP—Special Air Report

ALOHA—Aerial Locations of Hazardous Atmospheres

AMOS—Automated Meteorological Observing System

AOL—Alternate Operating Location

ARC—Acoustic Research Complex

ATC—Air Traffic Control

BWW—Basic Weather Watch

C—Celsius

CBRN—Chemical, Biological, Radiation, and Nuclear

CIG—Ceiling

CWW—Cooperative Weather Watch

DA—Density Altitude

DET—Detachment

DLT—Desired Lead Time

DO—Director of Operations

DOD—Department of Defense

ECC—Emergency Control Center

EDM—Effective Downwind Messages

EM—Emergency Management

EOC—Emergency Operations Center

ESF—Emergency Support Functions

FITS—Fighter Index of Thermal Stress

FLIPS—Flight Information Publications

HAFB—Holloman Air Force Base

HAFBI—Holloman Air Force Base Instruction

HRS—Hours

HSTT—High Speed Test Track

IAW—In accordance with

IDP—Installation Data Page

IFR—Instrument Flight Rules

IRC—Instrument Refresher Course

JET—Joint Environmental Toolkit

LFA—Local Flying Area

LGT—Light

LLWS—Low Level Wind Shear

LOX—Liquid Oxygen

MCC—Maintenance Control Center

MDT—Moderate

MEF—Mission Execution Forecast

METAR—Meteorological Aerodrome Report

METWATCH—Meteorological Watch

MGMT—Management

MISSIONWATCH—Mission Weather Watch

MSA—Munitions Storage Area

MX—Maintenance

MXS—Maintenance Supervisor

NBC—Nuclear, Biological, and Chemical

NDI—Non-Destructive Inspection

NM—Nautical Miles

NOTAMS—Notices to Airmen

NWS—National Weather Service

OPR—Office of Primary Responsibility

PACE—Primary, Alternate, Contingency, Emergency

PAR—Post-Attack Reconnaissance

PIREP—Pilot Report

PMSV—Pilot-to-Metro Service

POC—Point of Contact

POL—Petroleum, Oil, and Lubricants

PTD—Program Test Director

RAPCON—Radar Approach Control

RCR—Runway Condition Report

RDS—Records Disposition Schedule

RPA—Remotely Piloted Aircraft

RVR—Runway Visual Range

SFO—Senior Fire Officer

SLA—Service-Level Agreement

SOF—Supervisor of Flying

SVR—Severe

SWAP—Severe Weather Action Procedures

SWMT—Severe Weather Management Team

TAF—Terminal Aerodrome Forecast

TAWS—Target Acquisition Weather Software

URC—Unit Radar Committee

USAF—United States Air Force

USFS-WFAS—United States Forest Service Wildland Fire Assessment System

VCP—Volume Coverage Pattern

VFR—Visual Flight Rules

VIS—Visibility

WF—Weather Flight

WMO—World Meteorological Organization

WWA—Watches, Warnings, and Advisories

Terms

Aircraft Mishap—Term used to denote any event resulting in damage to, or destruction of, any aircraft, to include lightning strikes, inadvertent departure from the paved runway or taxiway surface, aircraft or Aerospace Ground Equipment (AGE) fires, and forced landings due to in- flight emergencies.

Amendment—Used as a message modifier when transmitting an aerodrome forecast amendment.

Climatology—Historical records of weather conditions measured or observed at a specific location is known as climatology. Some data goes back over 100 years, but generally, a 10 to 25 year history is more common. Climatology is useful in planning operations beyond 5 to 7 days. It usually describes the average (or mean) conditions such as high and low temperatures and extremes.

METWATCH—A deliberate process for monitoring the terrestrial weather or space environment in an area or region. The purpose of a METWATCH is to identify when and where observed

conditions significantly diverge from forecast conditions and determine courses of action to update or amend a forecast product or group of products and notify designated agencies.

Mission Execution Forecast (MEF)—Mission-tailored environmental information used in the execution of a mission. MEFs describe the meteorological mission environment and concentrate on environmental threats given specific operating thresholds. Where possible, courses of action to mitigate these threats are offered. WFs and WSTs conduct deliberate forecast processes to develop, deliver, monitor, and amend mission execution forecasts by fusing perishable data with operational and strategic level weather forecast products.

MISSIONWATCH—A deliberate process of monitoring terrestrial weather or the space environment for specific mission-limiting environmental factors that may adversely impact missions in execution. The MISSIONWATCH process is performed by WFs and WSTs and is intended to identify previously unidentified environmental threats and alert decision-makers at the operational unit and/or airborne mission commanders, enabling dynamic changes to mission profiles that may mitigate the environmental threat and optimize the chance of mission success

Severe Weather—Any weather condition that poses a hazard to property or life.

Attachment 2

FORMAT FOR WEATHER WATCHES, WARNINGS, AND ADVISORIES

Table A2.1. Forecast Weather Watch Criteria.

	Criteria	Text
Severe Weather Action Plan Criteria	Tornado	Potential for Tornado/Funnel Cloud exists A warning will be issued later if required.
	Damaging Winds	Potential for Damaging Winds exists greater than or equal to 50 knots. A warning will be issued later if required.
	Freezing Precipitation	Potential for Freezing Precipitation exists A warning will be issued later if required.
	Severe Thunderstorms	Potential for Severe Thunderstorms A warning will be issued later if required. Damaging Hail greater than or equal to 3/4 inch. and/or Damaging Winds greater than or equal to 50 knots.
Non-severe forecast watch criteria	Lightning within 5 NM	Potential for Lightning exists within 5 NM. A warning will be issued later if required.

Table A2.2. Forecast Weather Warning Criteria.

	Criteria	Text
Severe Weather Action Plan Warning Criteria	Tornado	Tornado within 5 NM. Take shelter immediately.
	Damaging Winds ≥ 50 kts	Damaging Winds greater than or equal to 50 kts. Not associated with thunderstorms forecast value XX kts.
	Severe Thunderstorms	Severe Thunderstorms. Damaging Hail greater than or equal to 3/4 inches. Forecast value XX inches and/or Damaging Winds greater than or equal to 50 kts forecast value XX kts.
	Freezing Precipitation	Freezing Precipitation (Any Intensity)
Non-Severe	Moderate Thunderstorms	Moderate Thunderstorms. Large Hail greater than or equal to 1/4 but less than 3/4 inches forecast value XX inches and/or Strong Winds

		greater than or equal to 35 but less than 50 kts forecast value XX kts.
	Blizzard Conditions	Blizzard lasting greater than or equal to 3 hours. Visibility less than or equal to 1/4 SM and Winds greater than or equal to 30 kts.
	Heavy Snow	Heavy Snow Accumulation greater than or equal to 2 inches within 12 hrs.
	Heavy Rain	Heavy Rain Accumulation greater than or equal to 2 inches within 12 hrs.
	Strong Winds ≥ 35 but < 50 knots	Strong Winds greater than or equal to 35 but less than 50 kts forecast value XX kts.

Table A2.3. Observed Weather Warning Criteria.

Criteria	Text
Lightning (Within 10 NM of the Test Track)	Observed Lightning within 10 NM of the Test Track.
Lightning (Within 5 NM)	Observed Lightning within 5 NM.

Table A2.4. Forecasted Weather Advisory Criteria.

Criteria	Text
Surface winds ≥ 25 but < 35 knots	Winds greater than or equal to 25 but less than 35 knots forecast value XX kts.

Table A2.5. Observed Weather Advisory Criteria.

Criteria	Text
Observed Blowing Dust	Observed blowing dust with visibility less than or equal to 5 SM.
Surface temperature > 32º F but < 40ºF	Observed Surface Temperature greater than 32º F but less than or equal to 40º F.
Surface temperature < 32º F	Observed Surface Temperature less than or equal to 32º F.
Thunderstorms within 25 NM of Holloman AFB	Observed Thunderstorms within 25 NM.
Thunderstorms within 10 NM of Holloman AFB	Observed Thunderstorms within 10 NM.
Ice FOD (temp ≥ -4º F and < 50º F and dew point within 5º F of the temp)	Observed ICE FOD Conditions.
Surface winds sustained ≥ 35 knots occurring in the LFA	Observed Surface Winds sustained 35 knots. Occurring in the Local Flying Area.
Light or greater icing in the LFA	Observed Icing Light or Greater in the Local Flying Area.
Moderate or greater turbulence in the LFA	Observed Moderate or Greater Turbulence in the Local Flying Area.
LLWS on station	Observed Low-Level Wind Shear.
Observed Crosswinds ≥ 15 but < 20 knots	Observed Crosswinds greater than or equal to 15 but less than 20 knots.
Observed Crosswinds ≥ 20 but < 25 knots	Observed Crosswinds greater than or equal to 20 but less than 25 knots.
Observed Crosswinds ≥ 25 knots	Observed Crosswinds greater than or equal to 25 knots.
Observed Surface Winds ≥ 31 knots	Observed Surface Winds greater than or equal to 31 knots. Occurring at Holloman AFB.

Table A2.6. Range Weather Watch Criteria.

Criteria	Text
Tornado	Potential for Tornado/Funnel Cloud exists. A warning will be issued later if required.
Damaging Winds	Potential for Damaging Winds exists greater than or equal to 45 knots. A warning will be issued later if required.
Severe Thunderstorms	Potential for Severe Thunderstorms A warning will be issued later if required. Damaging Hail greater than or equal to 3/4 inch. and/or Damaging Winds greater than or equal to 45 knots.
Lightning within 10 NM	Potential for Lightning exists within 10 NM. A warning will be issued later if required.

Table A2.7. Range Weather Warning Criteria.

Criteria	Text
Tornado	Tornado/Funnel Cloud expected
Damaging Winds	Damaging Winds greater than or equal to 45 kts. forecast value XX kts.
Severe Thunderstorms	Severe Thunderstorms. Damaging Hail greater than or equal to 3/4 inches. Forecast value XX inches and/or Damaging Winds greater than or equal to 45 kts forecast value XX kts.
Lightning within 10 NM	Observed Lightning within 10 NM.

Attachment 3

WEATHER IMPACT ON UNIT OPERATIONS

Table A3.1. F-16 Fighting Falcon.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
CIG/VIS < 3000 feet / 5 SM	Closes the Visual Flight Rules (VFR) traffic pattern.	1. Traffic pattern work rescheduled. 2. Change to IFR training. Impacts training of ATC personnel.
CIG/VIS < 1500 feet / 3 SM	Stops all VFR operations.	All aircraft require IFR sequencing.
CIG/VIS ≥ 200 feet / ½ SM to < 1500 feet / 3 SM or RVR 2400	Field IFR.	File IFR only.
CIG/VIS < 1500 feet / 3 SM	Stops Cat V flights.	Cat V pilots weather canceled/diverted.
CIG/VIS < 700 feet / 2 SM	Stops Cat IV flights.	1. Cat IV pilots weather canceled/diverted. 2. Cancels student flights.
CIG/VIS < 500 feet / 1½ SM	Stops Cat III flights.	Cat III pilots weather canceled/diverted.
CIG/VIS > 200 feet / ½ SM to < 300 ft / 1 SM	Stops Cat II flights and requires 54th OG/CC approval for Cat I flying operations. Normal training missions canceled.	1. Cat II pilots weather canceled/diverted. 2. Cancels instructor flights. 3. 54 FG Commander approval required to fly Cat I pilots.
CIG/VIS < 200 ft / ½ SM	Stops all flights	All rescheduled/diverted.
Crosswinds ≥ 25 kts	Cancels local aircraft operations	All rescheduled/diverted.
Crosswinds ≥ 20 kts	1. Wet runway: Instructor pilot Cannot land. 2. Dry runway: Student Cannot land.	Reschedule/divert.
Crosswinds ≥ 15 kts	Wet: Student cannot land	Reschedule/divert.
Sustained winds ≥ 25 kts	Cancels normal training	All Rescheduled.

over water	missions. Requires 54 FG Commander waiver for approval.	
Surface winds ≥ 25 kts	May cause aircraft damage	Close canopies, remove jack
Surface winds ≥ 35 kts to < 50 kts	1. Delays maintenance operations. 2. Cancels normal training missions in any area over 35 kts sustained. Requires 54 FG Commander waiver approval.	All rescheduled/diverted.
Surface winds ≥ 50 kts	Cancels aircraft operations. Control tower evacuated.	All rescheduled/diverted.
Thunderstorms in working area / ranges	Restrict flying in area / ranges.	Reschedule or adjust area for daily flying.
Lightning within 5 NM	No takeoffs. 54 FG Commander approval for landing. All maintenance personnel clear the airfield.	Suspend all airfield operations.
Hail ≥ ½ inch	Damage to aircraft.	1. Hangar aircraft. 2. Close canopies. 3. Evacuate personnel from flight line before hail starts.
Hail ≥ ⅛ < ½ inch	Damage to skin of aircraft.	1. Hangar aircraft. 2. Close canopies. 3. Evacuate personnel from flight line before hail starts.
Rainfall ≥ 2 inches in 12 hrs	Standing water on runways and taxiways.	Check with Airfield Management for Runway Condition Report (RCR).
Snowfall ≥ 2 inches in 12 hrs	Ice/snow on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
Freezing Precipitation ≥ Light	Icing on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
Moderate Icing	Changes aircraft flight characteristics.	Climb or descend only through icing area.
Severe Icing	Changes aircraft flight	Flight canceled.

	characteristics.	
< Moderate Turbulence	Bumpy ride.	Climb or descend through turbulence area.
≥ Moderate Turbulence	Aircraft control difficult.	Restrict flights.
Frost	Changes lift/drag ratio of airfoils.	De-ice aircraft

Table A3.2. T-38 Talon.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
CIG/VIS < 1500 feet / 3 SM	Stops all VFR operations.	All aircraft require IFR sequencing.
CIG/VIS ≥ 200 feet / ½ SM to < 3 SM or RVR 2400	Field IFR.	File IFR only.
CIG/VIS < 300 feet / 1 SM	Wing CC approval for flying operations. Normal training.	2. All rescheduled – 54 FG Commander approval required to fly.
CIG/VIS < 200 feet / ½ SM	Stops all flights.	All rescheduled/diverted.
Crosswinds ≥ 30 knots	Cancels local aircraft operations.	All rescheduled/diverted
Crosswinds ≥ 20 knots with wet runway	Cancels local aircraft operations.	All rescheduled/diverted.
Crosswinds ≥ 10 knots with ice / slush / snow on runway	Cancels local aircraft operations.	All rescheduled/diverted.
Surface winds ≥ 35 knots to < 50 knots	1. Delays maintenance operations. 2. Cancels normal training missions. Requires 49 OG Commander waiver approval.	All rescheduled/diverted.
Surface winds ≥ 50 kts	Cancels aircraft operations.	All rescheduled/diverted.
Thunderstorms in working area / ranges	Restrict flying in area / ranges	Reschedule or adjust area for daily flying.
Lightning within 5 NM	No takeoffs. 49 OG Commander or 704 Test Group Commander waiver approval to land. All	Suspend all airfield operations.

	maintenance personnel clear the airfield.	
Hail ≥ ½ inch	Damage to aircraft.	1. Hangar aircraft. 2. Close canopies. 3. Evacuate personnel from flight line before hail starts.
Hail ≥ ⅛ but less than < ½ inch	Damage to skin of T-38A.	1. Hangar aircraft. 2. Close canopies. 3. Evacuate personnel from flight line before hail starts.
Snowfall ≥ 2 inches in 6 hrs	Ice/snow on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
≥ Light freezing precip	Icing on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
Light rime icing	Changes aircraft flight characteristics.	Climb or descend only through icing area.
> Light rime icing	Changes aircraft flight characteristics.	Flight canceled.

Table A3.3. MQ-9 Reaper.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
CIG/VIS < 1500 feet / 3 SM	Increased recovery fuel needed.	Reschedule.
Crosswinds ≥ 15 knots	Generic maximum allowed limit; touch-and-go training max is 10 knots; asymmetric limit 13 knots.	Reschedule.
Tailwind ≥ 10 knots	Takeoff limit.	Reschedule.
Gust Spread > 20 knots	No takeoffs or landings	Reschedule.
Surface winds > 30 knots sustained	Impacts ground speed, must be avoided or mitigated as best as possible.	Reschedule.
Wet/Ice RCR ≤ 5*	No takeoffs or landings.	Reschedule.

Thunderstorms, rain, drizzle, or virga: few or greater in coverage which can't be avoided	• Avoid areas of TS/Rain, Drizzle or Virga • Cannot operate within 25 NM of thunderstorms.	Reschedule, recover, adjust operating area or obtain waiver.
Liquid, freezing, or frozen precipitation	No takeoffs or landings with hail, moderate rain, or freezing precipitation.	Reschedule.
≥ Light icing	No takeoffs or landings in known areas of icing.	Reschedule.
≥ 6500 feet Density Altitude (DA)	Causes lack of sufficient thrust; must reduce weight in higher DA values.	Reschedule.
≥ Moderate turbulence	May operate in forecast areas of turbulence, but not if observed.	Reschedule or adjust flying area.
Winds aloft ≥ 80 knots	Aircraft control compromised.	Reschedule or adjust flying area.
< 50% - broken ceiling	Cameras can't see below or through.	Reschedule or adjust flying area.
Thick dust or smoke	All cameras unusable.	Reschedule or adjust flying area.
Surface winds > 39G52 knots forecast	Cancels aircraft operations.	Reschedule.
< 40°F or > 95°F	Impacts launch/recovery; engine may be too cold or overheat.	Reschedule or adjust flying area.

Table A3.4. QF-16 drone.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
CIG/VIS < 3000 feet / 5 SM	Limits the VFR traffic pattern.	Refer to HAFBI 11-250.
CIG/VIS < 1500 feet / 3 SM	Stops all VFR operations Stops unmanned operations.	All manned aircraft require IFR sequencing.
CIG/VIS ≥ 300 feet / 1 SM to < 1500 feet / 3 SM	Field IFR.	File IFR only.
CIG/VIS < 300 feet / 1 SM	Stops Manned Flying.	Mission's weather canceled/diverted.
Sustained winds ≥ 15 knots	Stops unmanned operations.	Reschedule.
Crosswinds ≥ 25 knots All runways	Cancels local aircraft operations.	All rescheduled/diverted.
Surface winds ≥ 25 knots	Cancels normal training missions in any area over 35 kts sustained.	All rescheduled/diverted.
Surface winds ≥ 50 knots	Cancels aircraft maintenance operations.	All rescheduled/diverted.
Thunderstorms in working area / ranges	Restrict flying in area / ranges.	Reschedule or adjust area for daily flying.
Lightning within 5 NM	No takeoffs. SOF approval for landing. All maintenance personnel clear the airfield.	Suspend all airfield operations.
Hail ≥ ½ inch	Damage to aircraft.	1. Close canopies. 2. Evacuate personnel from flight line before hail starts.
Rainfall ≥ 2 inches in 12 hours	Standing water on runways and taxiways.	Check with Airfield Management for RCR.
Snowfall ≥ 2 inches in 12 hours	Ice/snow on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
Freezing precipitation / frost ≥ light	Icing on aircraft and runway condition changes.	De-ice and check with Airfield Management for RCR.
Moderate icing	Changes aircraft flight characteristics.	Minimize flight through icing area.

Severe icing	Changes aircraft flight characteristics.	Flight canceled.
≥ Moderate turbulence	Aircraft control difficult.	Restrict flights.

Table A3.5. UH-60 Blackhawk.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Temperature < -54° C or > +50° C	No takeoffs or landings.	All rescheduled/diverted.
Surface winds > 45 knots	No takeoffs or landings.	All rescheduled/diverted.
Thunderstorms	Avoid by 20 NM, restricts flying area, no takeoff or landing if at airfield.	Adjust flying area to avoid, divert.
Severe icing	No operations in area.	Reschedule or adjust flying area.
Severe turbulence	No operations in area.	Reschedule or adjust flying area.

Table A3.6. AFRL ARC Program Operations.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Winds ≥ 12 knots	No-Go	Reschedule test
Light precipitation	No-Go	Reschedule test
Visibility < 3 SM	No-Go	Reschedule test
Cloud ceiling < 3000 feet	No-Go	Reschedule test
Light turbulence	No-Go	Reschedule test
Light icing	No-Go	Reschedule test
Thunderstorms w/in 5 SM	No-Go	Reschedule test
Additional criteria dependent on test aircraft		

Table A3.7. Maintenance Activities.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Any weather watch		Increase awareness and plan for possible adverse weather conditions.
Weather warning of winds (gusts or sustained) ≥ 35 knots	Potential damage to aircraft.	1. Ensure all actions in previous weather conditions are complied with. 2. Hangar doors will remain closed except to remove/shelter aircraft, AGE or munitions, and during re/defueling operations. 3. Discontinue all egress maintenance outside of sheltered areas. 4. Remove all non-powered AGE from open ramp areas. Powered Aerospace Ground Equipment (AGE) will be removed from open ramp areas at the discretion of the MXS Superintendent. 5. Munitions up/down load may occur only on aircraft in authorized hangars with shelter doors closed. 6. Discontinue AGE refueling operations. 7. No outside munitions pickups or deliveries will be made without approval from 49 MXS/MXM or 49 MXS/MXMW.
Observed weather advisory of sustained winds > 35 kts	Stop hotpit refueling activities.	All aircraft refueling operations (Except in-shelter refueling) will be stopped upon MOC's verification of ACTUAL wind speed of sustained 35 kts or greater.

Weather warning of winds ≥ 50 kts or hail ≥ ¾ inch	Potential damage to aircraft.	1. Ensure all previous weather conditions are complied with. 2. Hangar as many aircraft as possible (especially if hail is forecast). The 49 WG aircraft will utilize 49 WG facilities to include the hush houses and fuel barns. 3. Park all remaining aircraft into the wind and moor. 4. Remove all miscellaneous items, to include AGE, from open ramp areas and terminate all outside aircraft maintenance. 5. Hangar doors will remain closed. 6. Terminate all outside munitions operations. 7. All aircraft refueling operations will be stopped.
Weather advisory of lightning within 25 NM	Potential damage to aircraft.	Be prepared to decrease maintenance activities.
Weather advisory of lightning within 10 NM	Potential damage to aircraft.	1. Terminate egress system maintenance outside sheltered areas. 2. Close canopies and secure panels on aircraft parked outside. 3. Terminate open fuel cell maintenance and secure access panels.
Weather warning of lightning within 5 NM	Potential damage to aircraft.	1. Evacuate personnel from locations containing explosives that could be initiated by lightning. Minimum required personnel may remain to carry out an urgent operational mission. 2. Discontinue all munitions actions. Within the Munitions

		Storage Area (MSA), the Munitions Section Chief or Munitions Control will order evacuation when considered necessary. 3. Discontinue all avionics and Non-Destructive Inspection (NDI) section operations. Power down all back shop avionics and NDI operations. 4. Personnel will be evacuated from these locations: 4.1. Explosive locations (excluding those licensed by AFMAN 91-201, *Explosive Safety Standards*, paragraph 2.35.), operating buildings, open storage sites, or loading docks without approved lightning protection systems, which contain explosives. 4.2. Facilities containing exposed explosives, explosive dust or vapor, or unpackaged electrically initiated explosive devices (even if equipped with approved lightning protection systems). 4.3. Parked explosive laden vehicles. 4.4. All locations within public traffic route of facilities and carriers listed above.

		5. When evacuating buildings containing explosives, staff operations requiring constant attention will be manned by the minimum number of personnel required until it is safe to leave. 6. Stop aircraft explosives loading, unloading, or preload operations at the same time fueling and defueling operations are suspended. 7. All fuel servicing and Petroleum, Oil, and Lubricants (POL) and/or fuel related maintenance activities, including Liquid Oxygen (LOX) servicing, will cease. 8. Terminate all vehicle and AGE fueling and defueling operations. 9. Ensure all personnel are off the open ramp areas. 10. NDI will power down X-Ray, Magnetic Particle Machine, and Oil Analysis Spectrometers.
Heavy snow or freezing precipitation warnings	Potential damage to aircraft.	1. Ensure all aircraft canopies and panels are closed. 2. Ensure all shelter doors are closed, except for removal/sheltering of aircraft, AGE, munitions, and re/defueling of aircraft. 3. All aircraft towing and vehicular movement will be held to a minimum. 4. Installed aircraft engine maintenance runs will be held to a minimum.

Table A3.8. 49 MSG.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Lightning within 5 NM warning	Threat to personnel.	Curtail outside activity and have personnel remain indoors until the weather advisory is lifted.
Lightning within 10 or 25 NM	Threat to personnel.	Consider limiting outside activities.
Hail	Threat to personnel and vehicle damage.	1. Move vehicles and equipment inside the stations to prevent possible damage. 2. Consider limiting outside activity to mission-essential actions.
Tornado Watch	Threat to personnel and vehicle damage.	1. Consider resource dispersal or sheltering of vehicles and equipment. 2. Consider restricting outside activities to mission-essential tasks only. 3. Establish 100% accountability of personnel. 4. Consider Emergency Communications Center (ECC) relocation during a "Watch" to reduce imminent danger to ECC Operators.
Tornado Warning	Threat to personnel and vehicle damage.	1. Alert personnel to immediately seek shelter. 2. Remain sheltered until proper authorities declare "All Clear". 3. Conduct a Post Attack Reconnaissance (PAR) report afterwards and report status to the ECC and SOF.

Winds ≥ 50 kts	Threat to personnel and vehicle damage.	Evacuate ECC to Fire Station #2.

Table A3.9. 49 MDG.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Severe Weather (Tornado, Winds ≥ 50 kts, or hail ≥ ¾ inch)	Threat to personnel.	1. Cease ambulance operations. 2. Activate Medical Control Center (MCC) upon 49 MDG/CC direction. 3. Stay away from windows and glass doors. 4. Limit outside movement 5. Move personnel to center of building.

Table A3.10. 49 SFS.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Lightning, tornado, winds ≥ 50 kts, or hail	Threat to personnel and vehicle damage.	1. Relocate boundary sentries to solid structure. 2. Mobile patrols park in sheltered area. 3. Wear PPE when out of shelter.

Table A3.11. Detachment 4, 2nd Weather Squadron.

CRITERIA	MISSION IMPACT	MITIGATION ACTION
Lightning, tornado, winds ≥ 50 kts, hail, or blowing dust	Threat to personnel and equipment.	1. Stow telescope.